

SECTION A: STRUCTURED QUESTIONS

QUESTION 1

The cell wall of the prokaryote *Escherichia coli* is made of peptidoglycan that consists of polysaccharide chains cross-linked by short chains of amino acids. **Fig. 1.1** shows that the polysaccharide chain is made up of two different sugars N-acetylmuramate (NAM) and N-acetylglucosamine (NAG), which alternate along the chain.

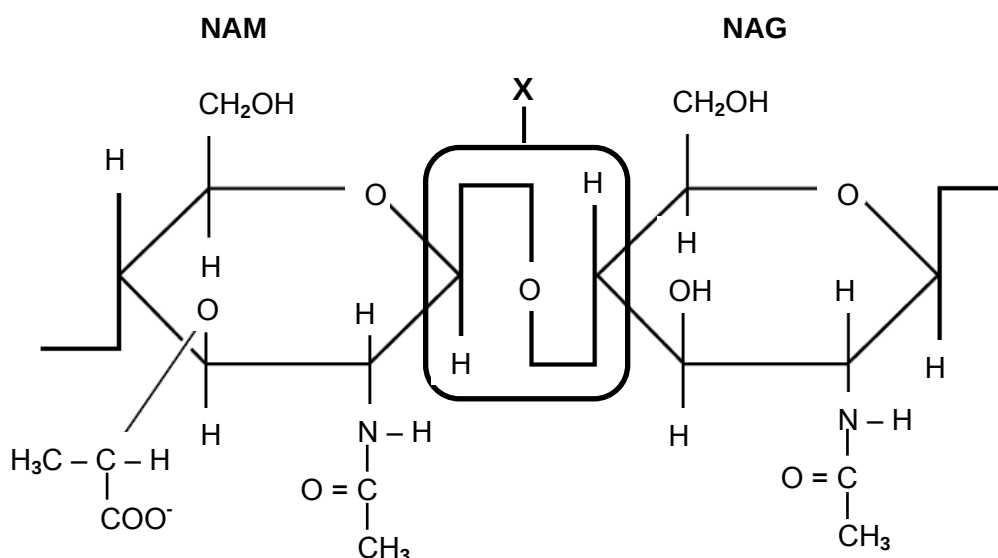


Fig. 1.1

- (a) State the name of the bond represented by X. [1]

- (b) Distinguish between bacterial peptidoglycan and cellulose found in the plant cell wall. [2]

- (c) Suggest why peptidoglycan is cross-linked with amino acids. [1]

Metabolic pathways can be controlled by end-product inhibition of enzyme-catalysed reactions. KAS III is the initial enzyme of fatty acid production in bacteria. The substrate for this reaction is malonyl-ACP.

Three different mutant strains of bacteria were generated, each with a different mutated *KAS III* gene: **M1**, **M2** and **M3**. The enzyme activity of the wild-type and the three mutant strains was tested with and without the addition of a KAS III inhibitor, dodecanoyl-ACP. Dodecanoyl-ACP is structurally similar to malonyl-ACP. **Fig. 1.2** shows the mean activity of KAS III in the wild-type, **M1**, **M2** and **M3** strains.

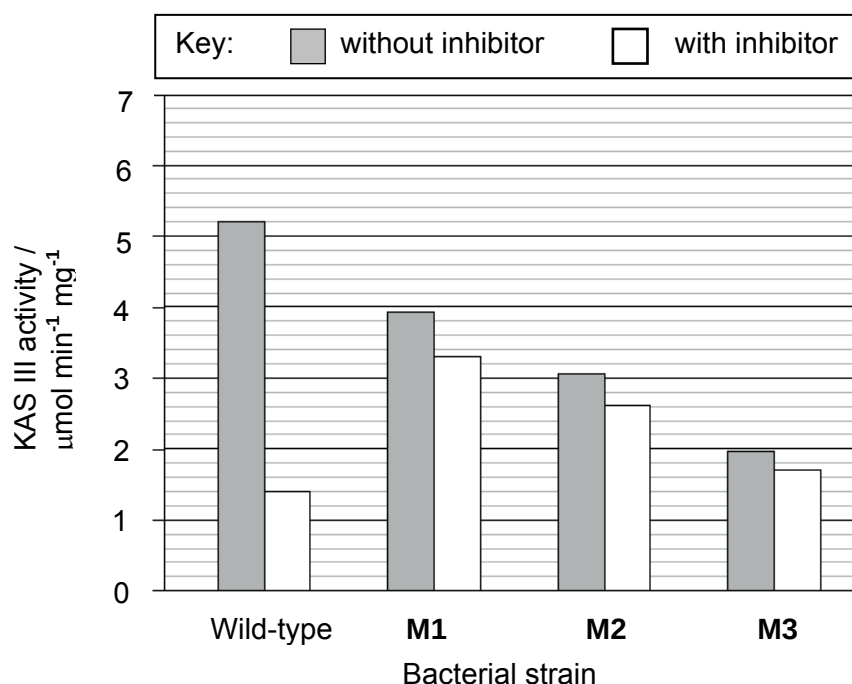


Fig. 1.2

- (d) With reference to **Fig. 1.2**, explain why the activity of KAS III in the wild-type strain changes when dodecanoyl-ACP is added. [2]

- (e) Suggest an explanation for the effect of mutation on KAS III activity. [2]

Fig. 1.3 shows the effect of increasing malonyl-ACP concentration on the rate of reaction catalysed by KAS III in the wild-type strain.

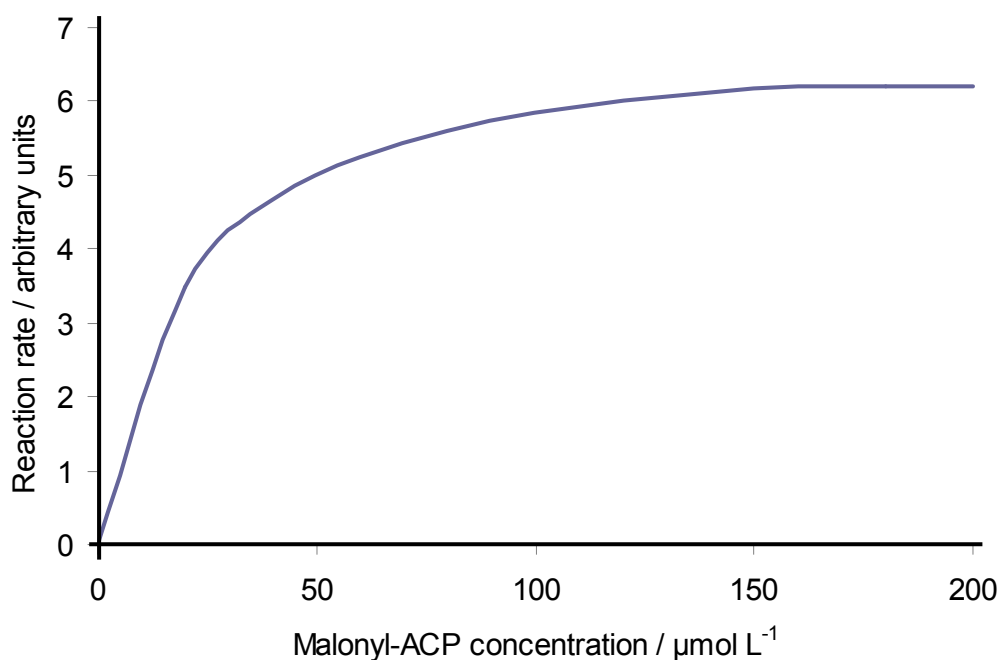


Fig. 1.3

- (f) Using information from **Fig. 1.2**, sketch a curve in **Fig. 1.3** to represent the effect of the addition of dodecanoyl-ACP on the rate of reaction with increasing malonyl-ACP concentration. [1]

[Total: 9 marks]

QUESTION 2

(a) Draw a fully labelled diagram of a 3-bp long DNA molecule, using the space provided below.

[3]

The DNA in the genome of an organism directs protein synthesis by using RNA as an intermediate, as shown in **Fig. 2.1**.

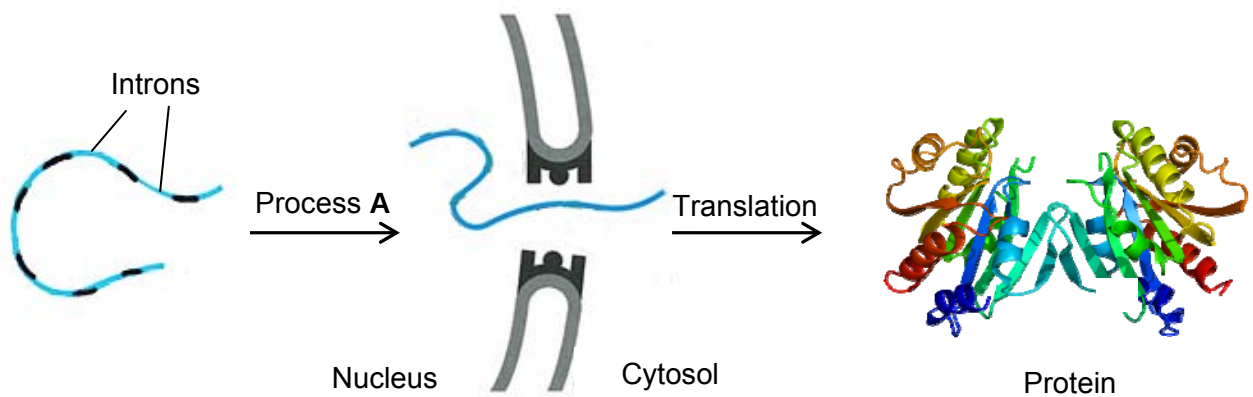


Fig. 2.1

(b) Describe briefly how RNA is synthesised from DNA.

[3]

(c) With reference to **Fig. 2.1**,

(i) name process **A**.

[1]

(ii) describe how process **A** allows for the formation of a functional mRNA molecule. [2]

(d) In addition to RNA, aminoacyl-tRNA synthetase is also required for protein synthesis to proceed. Explain how the structure of aminoacyl-tRNA synthetase allows it to perform its function. [2]

[Total: 11 marks]

QUESTION 3

Glucose-6-phosphate dehydrogenase (G6PD) deficiency is a sex-linked recessive condition.

- (a)** Define what is meant by a *sex-linked* condition. [1]

A normal Indian man marries a normal woman, whose father has G6PD deficiency. The couple decides to have a child.

- (b) (i)** State the probability that their first boy will be affected with G6PD deficiency. [1]

- (ii)** Explain your answer to **(b)(i)**. [2]

The man has blood group AB and the woman has blood group O. The ABO gene locus is located on chromosome 9.

- (c) Using suitable symbols, draw a genetic diagram in the space provided below to show the expected phenotypic ratio of the ability to produce G6PD and blood group in the offspring of the Indian couple. [5]

[Total: 9 marks]

QUESTION 4

Hepatitis B is an infectious illness caused by the hepatitis B virus. Vaccines, which have been developed for the prevention of hepatitis B viral infections, rely on the use of a viral envelope protein called hepatitis B surface antigen (HBsA). The bacterium *Escherichia coli* is commonly used to produce HBsA for the manufacture of vaccines.

Fig. 4.1 shows part of the hepatitis B viral genome which includes the *HBsA* gene together with the positions of the restriction sites of three restriction enzymes, *Eco*RI, *Hind*III and *Hpa*I. These restriction sites do not occur elsewhere within the hepatitis B viral genome.

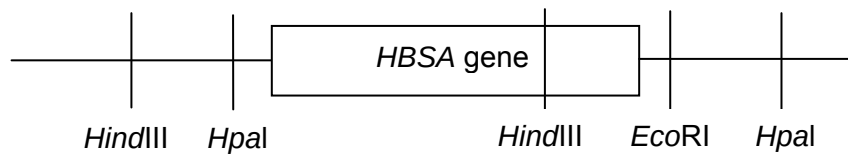


Fig. 4.1

Fig. 4.2 shows the bacterial plasmid pXMAS25 with the positions of the restriction sites of three restriction enzymes, *Eco*RI, *Hind*III and *Pst*I, available at its multiple cloning site. These restriction sites do not occur elsewhere within the plasmid pXMAS25.

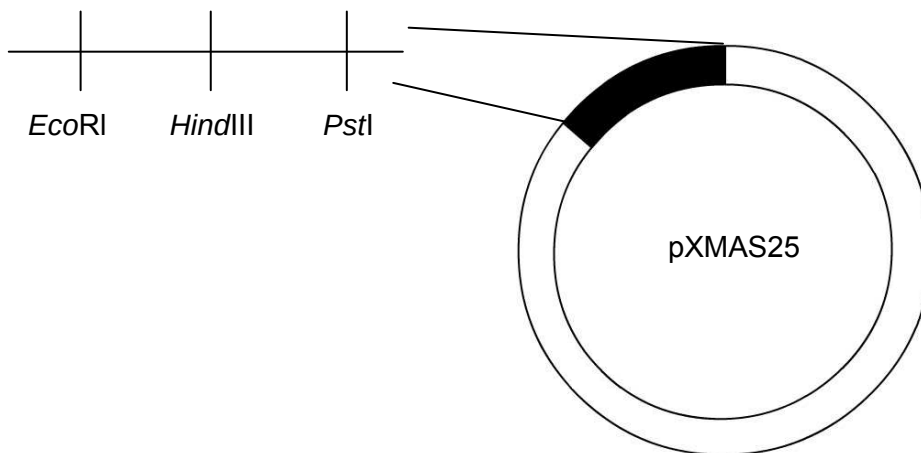


Fig. 4.2

(a) Name two important features of a plasmid not shown in pXMAS25 in **Fig 4.2**. [2]

All restriction sites in **Fig. 4.1** and **Fig. 4.2** generate sticky ends when cut, except for the *HpaI* site.

- (b) With reference to **Fig. 4.1** and **Fig. 4.2**, describe the procedure to create a recombinant pXMAS25 plasmid containing the viral *HBSA* gene. [5]

To produce recombinant eukaryotic proteins such as human growth hormone (hGH) using *E. coli* cells, the eukaryotic *hGH* gene cannot be obtained from genomic DNA, unlike the case of the viral *HBSA* gene.

- (c) Explain why this is so, and suggest how the *hGH* gene sequence can be derived from another source to ensure its proper expression in *E. coli*. [3]

In recent years, banana has been successfully engineered to produce HBSA using modern plant tissue culture techniques, incorporating the use of a modified plant virus to carry the HBSA gene into banana cells. By consuming these transgenic bananas, individuals will develop immunity against Hepatitis B. Apart from banana, potato is another common plant which can be engineered to produce HBSA. However, banana is the preferred choice for the production of oral vaccines against hepatitis B.

- (d) Suggest an advantage of using banana over potato to produce oral vaccines against hepatitis B. [1]

[Total: 11 marks]

--- END OF SECTION A ---

SECTION B: FREE RESPONSE QUESTIONS

Answer **ONE** of the following two free response questions.

Your answers must be in continuous prose, where appropriate.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

BEGIN EACH PART ON A FRESH SHEET OF WRITING PAPER.

A **NIL RETURN** is required.

QUESTION 5

- (a) Describe the role of substrate-level phosphorylation in the production of ATP. [6]
- (b) With reference to photosynthesis and cellular respiration, explain the importance of compartmentalisation within a plant cell. [10]
- (c) Suggest the effects of a season of drought on the process of Calvin cycle in a plant. [4]

QUESTION 6

- (a) Describe the changes involving chromosomes during mitosis, and explain how these changes ensure genetic stability. [8]
- (b) Describe the fluid mosaic model of membrane structure and its significance to the nucleus. [8]
- (c) Suggest if the introduction of *Bt* corn into a wild corn population would result in evolution. [4]

[Total: 20 marks]

--- END OF SECTION B ---

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